

Gassand physics glossary

Every mathematical symbol from the AP-Physics-B/C treatment of the helium/sand zero-pressure balloon, with the value it takes in this sim (sources: `js/config.js` `DEFAULT_GASSAND`, `js/balloon_gassand.js`, and the measured 49 m/kg settling probe).

Symbol	Name / meaning	Units	Value in this sim
W	weight of the whole system, mg	N	$\approx 139 \times 9.81 \approx 1366$ at launch
F_B	buoyant force (Archimedes)	N	equals W at the float; code computes net $(\text{displacedAirMass} - \text{totalMass}) \cdot G$
F_D	drag force, $-\frac{1}{2}C_d\rho A v v$	N	sign opposes motion (<code>balloon_gassand.js:153</code>)
F_{net}	net vertical force fed into $m dv/dt$	N	buoyancy + drag
ΔF	buoyancy imbalance right after a release	N	$\approx g \Delta m_{\text{sand}}$ for sand; $\approx 6.236 g \Delta m_{\text{He}}$ for helium (release ladders paired by this factor)
m, m_{total}	total mass: structure + sand + helium	kg	$100 + 20 + 19.24 \approx 139.24$ at launch (<code>DRY_MASS_KG</code> + reserves)
m_{He}	helium mass aboard (only depletes)	kg	19.24165 at launch (<code>HELIUM_INIT_KG</code>)
Δm	mass shed by one release	kg	sand actions 0.005–1.0; helium actions 0.0008–0.1604
g (code <code>G</code>)	gravitational acceleration	m/s ²	9.81
v (code <code>vv_m_s</code>)	vertical velocity, + up	m/s	clamped to ± 2.5 (<code>MAX_VV_M_S</code>)
v_{terminal}	drag-limited vertical speed, $\sqrt{2\Delta F/(\rho C_d A)}$	m/s	0.11–1.56 across the action ladder; $\propto \sqrt{\Delta m}$, so a $200\times$ mass range gives only $\sim 14\times$ speed
M_{air}	molar mass of air	kg/mol	0.0289647 (<code>M_AIR_KG_MOL</code>)
M_{He}	molar mass of helium	kg/mol	0.0040026 (<code>M_HE_KG_MOL</code>)
$M_{\text{air}}/M_{\text{He}}$	air displaced per kg of helium (bubble regime)	—	7.237 (<code>AIR_PER_HELIUM</code>); net lift per kg He = 6.236
R	universal gas constant, in $V = m_{\text{He}}RT/(M_{\text{He}}P)$	J/(mol K)	8.314
T	ambient temperature (ideal-gas law contexts)	K	from <code>atmosphereAt(alt)</code>
P	ambient pressure	Pa	from <code>atmosphereAt(alt)</code>
$\rho, \rho_{\text{air}}(h)$	ambient air density at altitude h	kg/m ³	≈ 0.139 at the 17.1 km float ($= m_{\text{total}}/V_{\text{env}}$)
ρ_0	sea-level reference density	kg/m ³	≈ 1.225
H	density scale height in $\rho(h) = \rho_0 e^{-h/H}$	m	≈ 6800 near these altitudes (fit to the sim atmosphere, not a config constant)
$V, V_{\text{displaced}}$	volume of air displaced by the gas bubble	m ³	$\min(\text{bubble}, \text{envelope})$ — the two-regime <code>min</code> in <code>displacedAirMass</code>
V_{env}	envelope volume (fixed)	m ³	1000 (<code>V_ENVELOPE_M3</code>)
r	balloon radius (sphere of volume V_{env})	m	$\sqrt[3]{3V_{\text{env}}/4\pi} \approx 6.20$ (<code>balloonRadius_m</code>)
A	cross-sectional area for drag, πr^2	m ²	≈ 121
C_d	drag coefficient	—	0.47, sphere (<code>DRAG_COEFFICIENT</code>)
h	altitude	m	sim band 15,000–22,000 (<code>ALT_MIN_M/ALT_MAX_M</code>)
h_{park}	park (float) altitude solving $\rho(h) V_{\text{env}} = m_{\text{total}}$	m	17,093 at launch config (<code>altBandLow_m</code>); 18,076 with all sand dropped
Δh	shift in park altitude per mass shed, $\approx H(\Delta m/m)$	m	≈ 49 per kg of sand (measured); 983 for the whole 20 kg bag
k	effective spring constant of the float, mg/H	N/m	$139 \cdot 9.81/6800 \approx 0.20$ — matches the code comment “restoring force ~ 0.2 N/m”
ω	natural angular frequency of the float, $\sqrt{g/H}$	rad/s	≈ 0.038
T_{osc}	vertical oscillation period, $2\pi/\omega$ (distinct from temperature T)	s	≈ 166 (~ 2.8 min) — why the integrator sub-steps at 5 s (<code>VERT_SUBSTEP_S</code> , stability needs $dt \cdot \omega < 2$)

Two symbols to keep un-confused: T is temperature in the ideal-gas law but a period in the oscillation analysis

(written T_{osc} here), and ρ always means *ambient air* density — the helium’s own density never appears because the bubble-regime buoyancy $m_{\text{He}} (M_{\text{air}}/M_{\text{He}}) g$ already folds it in.